DCR-RAG: Draft-Conditioned Ranking with Conservative Passage Replacement

Anonymous Author(s)

Abstract

Passage selection in retrieval-augmented generation should supply a compact context that supports accurate answers. However, independent ranking by question relevance does not fully reflect a passage's value to the reader, and highly scored passages may repeat information while omitting key facts needed to connect a multi-hop answer. We propose DCR-RAG, a passage selection framework that combines draft-conditioned ranking with conservative pair selection. The framework generates a short candidate-pool answer draft and conditions a trainable passage scorer on it together with the question. It then constructs bounded single-passage replacements around the scorer's two selected passages and uses an ensemble of gain predictors to estimate each proposal's reader benefit relative to the original pair. The selector controls replacement decisions through a prediction-disagreement penalty and a gain threshold, retaining the original pair when the expected gain is insufficient. The reader finally generates an answer from the question and selected passages. Across five question-answering benchmarks, DCR-RAG achieves 54.46 macro F1 under a common two-passage budget, outperforming the evaluated SetR configuration by 2.82 percentage points. With the learned scorer fixed, conservative pair selection contributes an additional 0.92 F1 points over independent Top-2 selection.

CCS Concepts

• **Information systems** → **Retrieval models and ranking**; • **Computing methodologies** → *Natural language generation.*

Keywords

Retrieval-augmented generation, passage selection, draft conditioning, reader-aware reranking

ACM Reference Format:

Anonymous Author(s). 2027. DCR-RAG: Draft-Conditioned Ranking with Conservative Passage Replacement. In *Proceedings of the ACM Web Conference 2027 (WWW '27), May 10–14, 2027, Dublin, Ireland*. ACM, New York, NY, USA, 12 pages.

1 Introduction

In Web search and conversational information access, retrieval-augmented generation (RAG) grounds answers in externally retrieved content. Under limited reader context budgets, a system must select a compact set of complementary passages from retrieved candidates. However, individually relevant search results may repeat information while omitting the entity–relation connections needed for complete multi-hop answers. Effective passage selection must therefore consider not only individual passage relevance but also how passages jointly support accurate answer generation.

The difficulty is that passage relevance to a question does not fully reflect its contribution to answer generation. A passage may mention the correct entity without providing the relation the question asks about; two individually relevant passages may also repeat the same information while omitting a key fact needed to connect the answer. For example, answering "Which university did the author of a work graduate from?" requires both the work–author relation and the author's educational background. If both highly scored passages focus on the work and its author, they cannot provide complete support for the answer, despite their relevance to the question. Under a two-passage budget, one redundant passage can occupy the slot needed for the connecting fact. Passage selection must therefore address two related problems: estimating each passage's utility to the reader and identifying passage pairs that jointly provide complementary information.

Prior work already connects passage selection to reader behavior. GainRAG learns passage preferences from generation-side gain supervision [12]; SetR selects complementary passages through query information requirements [18]; R-CPS reranks and groups passages using reader answer distributions [35]. DCR-RAG builds on these views with two specific choices. First, a short answer draft generated from the retrieved pool conditions each trainable passage score together with the question and candidate passage. Second, bounded replacements of the independently scored Top-2 pair are accepted according to predicted reader gains over that pair. GAR, HyDE, and query rewriting also enrich retrieval inputs with generated text [9, 23, 24]. Here the draft helps assess original candidates without occupying a passage slot. We ask what its content adds beyond scorer learning and whether a conservative replacement rule improves on retaining the initial pair.

Our central idea is to use the answer draft as context for passage scoring and formulate pair adjustment as a gain decision relative to the current passage pair. A short pool-generated draft explicitly presents a tentative answer and its associated entities and relations, giving the scorer clues about answer construction beyond the question. Even an incomplete draft may help the scorer identify passages that supplement or support the answer. A draft can also be wrong and bias scoring toward passages that reinforce the incorrect answer. Subsequent adjustments should therefore consider their expected effect on the final reader and retain the existing pair when the predictions provide insufficient grounds for replacement.

We propose DCR-RAG, a passage selection framework combining draft-conditioned scoring with conservative pair selection. A frozen reference generator first produces a short answer draft from the candidate pool. A trainable scorer jointly takes the question, draft, and each candidate passage to form an independent Top-2 passage pair. The framework then constructs bounded single-passage replacements, each retaining one original passage. An independently trained ensemble of reader-gain predictors estimates their

WWW '27, Dublin, Ireland
2027.

answer-F1 improvement over the initial pair. The selector penalizes disagreement among predictors and adopts a replacement only when the adjusted predicted gain exceeds an acceptance threshold; otherwise it falls back to the initial pair. The final reader generates its answer from the question and two selected passages from the original pool, without receiving the draft. Gold answers are used only for offline supervision and evaluation; inference-time selection uses the question, candidate pool, draft, and available features.

We evaluate DCR-RAG on five question-answering datasets, comparing systems with shared candidate pools, a common final reader, and a two-passage budget. DCR-RAG achieves 54.46 macro F1, exceeding the evaluated SetR configuration by 2.82 percentage points. Held-out confirmation using the main checkpoints evaluates the complete system on disjoint questions. Fixed-draft component comparisons distinguish scorer learning from the added selector gain. Matched no-draft retraining tests draft access, while a separate equal-length query-expansion control tests draft content. Acceptance and harm counts characterize the selector's decision trade-off. Analyses of erroneous drafts, passage budgets, and complete request costs characterize the method's limits on deeper multi-hop questions and its quality–latency tradeoffs.

Our main contributions are:

- **Pool-conditioned draft-guided passage scoring.** We use a short answer draft generated from the retrieved candidate pool to condition passage scoring on tentative answer information and entity–relation clues. The draft guides passage selection without occupying the final reader's passage budget.
- **Local relative-gain passage selection.** We formulate pair adjustment as bounded single-passage replacement relative to an initial Top-2 pair. Direct reader-gain prediction, disagreement-aware acceptance, and fallback to the original pair jointly determine whether a replacement is adopted.
- **Controlled evaluation of both mechanisms.** Draft-source and teacher ablations evaluate pool-conditioned scoring, while fixed-scorer comparisons, alternative gain formulations, and replacement-risk analyses characterize the effectiveness and limitations of relative-gain selection.

2 Related Work

2.1 Reader-Oriented Retrieval and Reranking

DrQA, DPR, and ColBERT establish retrieval pipelines for open-domain answering and passage search [5, 15, 16]. Prior work aligns retrieval and reading through answer supervision or language-model objectives [10, 19, 20], and uses reader feedback to learn passage preferences, relevance judgments, and information-use capabilities [8, 32, 37]. GainRAG estimates the gain a passage provides for correct outputs and trains a preference-alignment module linking retrieval and generation models [12]. Other work optimizes RAG rerankers using language-model feedback and reinforcement learning [27, 34].

Large language models also perform or learn reranking directly. RankGPT explores listwise reranking with language models [28]; GenRT jointly models reranking and context truncation [36]; and REARANK, Rank-R1, and ERank explore reasoning, supervised fine-tuning, and reinforcement learning for reranking [3, 39, 40]. GainRAG learns question–passage preferences from gain signals; DCR-RAG jointly conditions each learned score on the question, pool-generated draft, and passage. A separate pair predictor learns reader-F1 changes for accepting or declining replacements.

2.2 Generated Context for Retrieval and Ranking

Iterative retrieval combines search with multi-step reasoning and generation feedback [1, 13, 30, 31]. Generated text can supply answer clues and semantic information beyond the original question to improve retrieval inputs. GAR expands retrieval queries with generated context [24]; HyDE generates a hypothetical document from the question and uses its representation for dense retrieval [9]; and query rewriting transforms the original question into an expression better suited to retrieval [23]. They differ in the form of the generated content, the information available during generation, and where that content enters the retrieval pipeline.

Speculative RAG generates multiple drafts from distinct retrieved subsets using a smaller specialist model and verifies them with a larger model [33]. DCR-RAG instead uses a pool-conditioned draft to guide passage scoring and selection; the final reader receives only the selected original passages.

DCR-RAG conditions passage scoring on the question, a short pool-generated draft, and the candidate passage, using tentative answers and entity/relation clues to assess utility. GainRAG adds a question-generated pseudo-passage to its candidate set [12]; our draft conditions scoring and occupies no final passage slot.

2.3 Passage Set Selection and Risk Control

Passage set selection examines whether multiple passages jointly satisfy a question's information requirements. Independently ranked passages may repeat information, so set quality also depends on their complementarity. SetR reasons about the question's information requirements and selects passages that jointly satisfy them [18]. R-CPS guides passage selection using reader answer distributions and groups passages by their predicted answers to select consistent passages [35]. DCR-RAG starts from an independent Top-2 pair and predicts whether a bounded single-passage replacement improves its reader outcome, with an explicit option to retain that pair.

Passage selection must also account for ineffective or misleading information. Risk-aware reranking adjusts candidate rankings using signals such as information gain [22], while counterfactual and critic mechanisms provide feedback for passage assessment and answer correction [7, 21, 25]. Within a fixed candidate pool, DCR-RAG makes bounded adjustments to the independent Top-2 pair. Each proposal retains one initial passage and uses its observed reader-F1 difference from the initial pair as the offline target. At inference, predicted gain, disagreement, and a threshold determine whether to replace or retain that pair. The design links draft-conditioned passage scoring to an explicit decision about changing the selected passage pair.

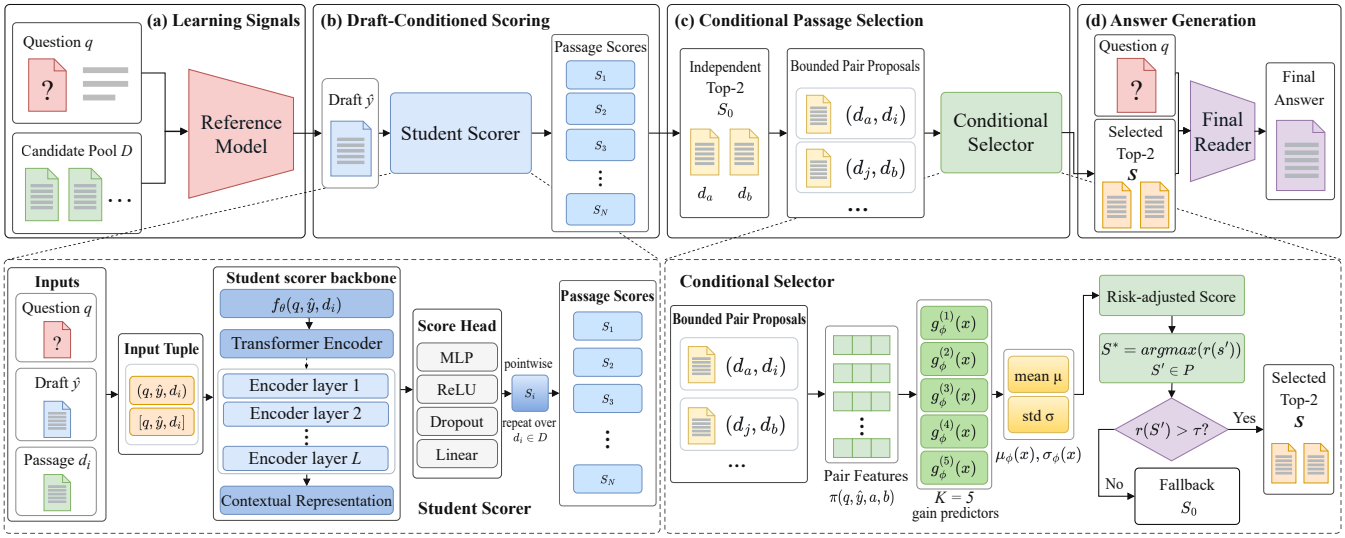


Figure 1: DCR-RAG: (a) offline learning signals and draft generation, (b) passage scoring, (c) conditional pair selection, and (d) final reading. Lower panels expand the scorer and five-predictor selector. The final reader sees original passages, without the draft.

3 Methodology

3.1 Task and Framework Overview

Given a question q and a fixed candidate pool $D = \{d_1, \ldots, d_N\}$, DCR-RAG selects two original passages for a final reader G. Retrieval precedes selection. The passage budget is $|S| = 2$; the objective is the quality of the answer $y = G(q, S)$, rather than the independent relevance of each passage. The pool contains at least two candidates in our evaluation.

Figure 1 separates offline supervision from inference. A frozen G_{ref} generates draft $\hat{y}$ from (q, D); scorer f_θ uses question, draft, and passage to form independent Top-2 S_0. Bounded proposals $\mathcal{P}(q)$ replace one passage at a time. Five gain predictors, a disagreement penalty, and a threshold decide acceptance. The final reader sees only the question and original passages; the draft occupies no passage slot.

The two learned components have different targets. Passage scoring learns which candidates the teacher prefers, while pair prediction learns how a replacement changes the final reader's F1. Teacher outputs and observed reader gains construct supervision offline. At inference, selection uses the draft, pool text, and fixed models; no gold answer or observed candidate-answer F1 enters the decision.

3.2 Draft-Conditioned Passage Scoring

The reference generator produces

$$\hat{y} = G_{\text{ref}}(q, D). \tag{1}$$

The frozen Qwen2.5-3B-Instruct generates the draft greedily from passage titles and text in BGE order, with 4,072 input tokens, right truncation, and at most 24 output tokens. Each question's draft is generated once and reused for teacher scoring and passage selection. Its tentative answer and entity/relation clues condition passage scoring; the final reader receives only original passages.

The scorer follows the lower Student Scorer panel in Figure 1. Question and draft form the question–draft field, and one candidate passage forms the passage field. The BGE-reranker-base tokenizer jointly encodes the fields. The Transformer encoder supplies contextual representations to the passage score head, which outputs

$$s_i = f_\theta(q, \hat{y}, d_i), \qquad S_0 = \text{Top2}_{d_i \in D}\, s_i. \tag{2}$$

All candidates share the question and draft. The two highest-scoring IDs define the baseline pair; ties follow BGE order.

The main teacher combines frozen BGE scores b_i with deletion responses N_i: the positive decrease in the fixed draft's mean log likelihood after removing d_i, evaluated by the same 3B reference. Its utility is

$$u_T(q, d_i) = z_D(b_i) + 0.75 \min(N_i/c, 1), \tag{3}$$

where z_D is pool min–max normalization and c is the 90th percentile of positive responses in that dataset's scorer-training set. The two highest utilities form T_q (Appendix A.2). We form selected–unselected preferences

$$\mathcal{B}_q = \{(i, j) : d_i \in T_q,\ d_j \in D \setminus T_q\} \tag{4}$$

and optimize the equal-weight logistic loss

$$\mathcal{L}_{\text{score}} = -\frac{1}{|\mathcal{B}_q|} \sum_{(i,j) \in \mathcal{B}_q} \log \text{sigmoid}(s_i - s_j). \tag{5}$$

The loss uses only Top-2 membership: positive global rescaling of teacher utility leaves its labels unchanged. Main preference labels use no gold answers. A separate 7B-reference control compares Scalar, Necessity, and Structured teachers. The deployed scorer is fixed.

Algorithm 1 DCR-RAG inference

Require: Question q, pool D; fixed models and configuration
$\hat{y} \leftarrow G_{\text{ref}}(q, D)$
$s_i \leftarrow f_\theta(q, \hat{y}, d_i)$ for every $d_i \in D$
$S_0 \leftarrow \text{Top2}_{d_i \in D}\, s_i$; $S \leftarrow S_0$
$\mathcal{P}(q) \leftarrow$ deduplicated single-passage replacements
if $\mathcal{P}(q) \neq \emptyset$ **then**
  for each $S' = \{d_u, d_v\} \in \mathcal{P}(q)$ **do**
    $x_{S'} \leftarrow \pi(q, \hat{y}, u, v)$; compute $\mu(S'), \sigma(S')$
    $r(S') \leftarrow \mu(S') - \kappa\sigma(S')$
  end for
  $S^* \leftarrow \arg\max_{S' \in \mathcal{P}(q)} r(S')$
  if $r(S^*) > \tau$ **then**
    $S \leftarrow S^*$
  end if
end if
Display S in original BGE order; $y \leftarrow G(q, S)$
return S, y

3.3 Conditional Proposals and Pair Features

Let $S_0 = \{d_a, d_b\}$. For each position, consider the four highest-scoring admissible replacement passages, retaining the other initial passage. The alternatives have the form $\{d_a, d_i\}$ or $\{d_j, d_b\}$, contain distinct passage IDs, and differ from S_0. Removing duplicate pairs yields $\mathcal{P}(q)$, with at most eight slots before deduplication. If too few candidates are available, only available replacements are considered. An empty proposal set returns S_0.

Retaining one initial passage bounds search and preserves part of the initial context, but limits recovery if both passages are unsuitable. Eight slots define a local search rather than optimization over all pairs.

For $S' = \{d_u, d_v\}$, the feature map is

$$x_{S'} = \pi(q, \hat{y}, u, v). \tag{6}$$

Features include normalized student and BGE scores, question coverage, residual coverage, passage overlap and length, and draft agreement. Passage scores convey individual ranking preferences. Coverage and overlap describe whether the pair adds question-related content or repeats it; draft agreement connects these signals to the tentative answer. For question token set Q and passage token sets T_u, T_v,

$$\text{cover}(u, v) = \frac{|Q \cap (T_u \cup T_v)|}{\max(1, |Q|)}, \tag{7}$$

$$\text{residual}(v \mid u) = \frac{|(Q \setminus T_u) \cap T_v|}{\max(1, |Q \setminus T_u|)}. \tag{8}$$

The residual term measures question words supplied by d_v after accounting for d_u. These are lexical features of available text, separate from the reader-F1 target used to train the predictor.

Feature order is shared across datasets. Fitted normalizers and coefficients remain fixed with each configuration. Standardization uses the fitting split and is frozen at inference. Appendix A.1 records feature handling. The initial pair defines proposals, the gain target, and fallback, but is not separately encoded in $x_{S'}$. Section 5.4 compares explicit initial-pair alternatives.

3.4 Learning Reader Gains

Pair prediction is trained separately from passage scoring. For each training question, the same final reader answers from the initial pair and each proposed pair, under the common passage budget and decoding settings. Scoring their complete answers against gold aliases gives

$$\Delta(S') = F_1\big(G(q, S')\big) - F_1\big(G(q, S_0)\big). \tag{9}$$

For fixed (q, S_0), true gain and absolute F1 rank proposals identically: the initial pair's F1 is constant. Acceptance additionally requires estimating improvement over that pair. Ranking and replacement acceptance are distinct tasks, with zero gain as the reference.

Each dataset supplies 100 gain-training questions and approximately 700 labeled proposals. Five deterministic folds group proposals by question, so alternatives from one question remain together. For fold k, ridge predictor $g_\phi^{(k)}$ fits standardized features on the other four folds with regularization coefficient $\alpha = 10$. These five fits are retained for inference. Scorer training uses a separate, disjoint set of 100 questions per dataset. Appendix A.2 gives the fold assignment.

All $K = 5$ predictors evaluate a proposed pair. Their mean estimates its gain and their population SD describes disagreement:

$$\mu(S') = \frac{1}{K}\sum_{k=1}^{K} g_\phi^{(k)}(x_{S'}), \tag{10}$$

$$\sigma(S') = \sqrt{\frac{1}{K}\sum_{k=1}^{K}\big(g_\phi^{(k)}(x_{S'}) - \mu(S')\big)^2}. \tag{11}$$

The divisor is K (ddof $= 0$), distinct from experimental sample SD across training seeds. Folds overlap; disagreement penalizes decisions without calibrating failure probability.

3.5 Conservative Selection and Answer Generation

The selector adjusts predicted gain by disagreement:

$$r(S') = \mu(S') - \kappa\sigma(S'), \qquad \kappa \geq 0. \tag{12}$$

For nonempty $\mathcal{P}(q)$, it finds the highest-scoring proposal and applies the acceptance threshold:

$$S^* = \arg\max_{S' \in \mathcal{P}(q)} r(S'), \tag{13}$$

$$S = \begin{cases} S^*, & r(S^*) > \tau, \\ S_0, & \text{otherwise.} \end{cases} \tag{14}$$

Accept only when the best disagreement-adjusted gain exceeds τ; otherwise retain S_0. Development-selected κ, τ, weights, and normalizers can differ by dataset; acceptance and fallback follow the same rule.

Algorithm 1 summarizes inference. Both scoring routes pass two original passages in BGE order to the common reader, without the draft.

For N candidates, passage scoring makes N evaluations; selection uses five predictors on at most eight proposals, versus $N(N-1)/2$ exhaustive pairs. Complete-request timing includes draft and answer generation.

Table 1: Main-test EM/F1 (%) on 4,500 questions with common pools, reader, and Top-2 passages. Macro weights five datasets equally; EM and F1 maxima are bold. Last column: DCR-RAG minus method macro F1 and paired question-bootstrap 95% CI; dash: unavailable or inapplicable.

Method	HotpotQA	2Wiki	MuSiQue	NQ	TriviaQA	Macro EM/F1	DCR-RAG minus method ΔF_1 [95% CI]
BGE Top-2	62.40/72.71	23.90/38.83	9.80/19.87	23.50/38.34	53.10/64.78	34.54/46.90	+7.56 [6.57, 8.55]
BGE+MMR	61.80/72.28	23.20/37.76	10.00/19.98	23.80/38.66	53.50/65.24	34.46/46.78	+7.68 [6.70, 8.66]
BGE + same draft	64.60/75.16	26.00/42.01	12.60/23.21	24.50/39.41	53.30/65.80	36.20/49.11	+5.35 [4.42, 6.28]
GainRAG-controlled	55.80/65.93	21.60/35.90	12.60/20.92	27.60/42.69	53.80/65.36	34.28/46.16	+8.30 [7.29, 9.31]
REARANK-7B	62.70/73.58	33.00/50.16	13.00/24.23	28.10/42.85	55.40/66.85	38.44/51.53	+2.93 [2.04, 3.82]
ERank-4B	61.00/71.68	23.90/38.61	13.80/24.87	**30.50/45.80**	**56.10**/68.98	37.06/49.99	+4.47 [3.53, 5.41]
Rank-R1-3B	54.80/64.59	22.30/37.42	11.40/21.11	27.80/42.59	51.50/64.59	33.56/46.06	+8.40 [7.36, 9.44]
SetR-2 (controlled)	66.00/76.70	28.90/45.62	15.40/26.12	28.60/43.32	54.30/66.43	38.64/51.64	+2.82 [1.91, 3.74]
DCR-RAG	**66.20/76.82**	**33.80/52.06**	**16.80/30.08**	28.60/44.18	56.00/**69.16**	**40.28/54.46**	—

4 Experimental Setup

4.1 Data, Candidate Pools, and Splits

We use HotpotQA [38], 2WikiMultiHopQA [11], MuSiQue [29], Natural Questions (NQ) [17], and TriviaQA [14]: 1,000 test questions each except 500 MuSiQue, totaling 4,500. Fixed pools have 2–10 HotpotQA passages, 17–20 MuSiQue, and 10 elsewhere; comparisons share IDs, text, and BGE order. Hash-ordered tests exclude training, development, model-selection, and relevant confirmation questions, checked by ID/normalized text. Separate confirmation uses 1,000 questions each on HotpotQA/2Wiki. The evaluated MuSiQue subset contains 24 two-hop, 324 three-hop, and 152 four-hop questions and is not representative of the original dataset's hop distribution. Under the Top-2 passage budget, complete support coverage is possible only for the two-hop subset.

4.2 Models and Comparison Protocols

The reference/reader are Qwen2.5-3B/7B-Instruct [26]; the scorer starts from BGE-reranker-base (XLM-R) [2, 6]. Common-pool Top-2 baselines use the same reader: BGE, BGE+MMR [4], unmodified BGE+draft, GainRAG-controlled [12], REARANK-7B (listwise weights) [39], ERank-4B (pointwise weights) [3], Rank-R1-3B (released adapter) [40], and SetR (Llama-3.1-8B-Instruct) [18]. GainRAG-controlled is fine-tuned per dataset and omits pseudo-passages. SetR selects a fixed-cardinality set under the shared budget, displayed in BGE order, with a 10,240-token selector context (Appendix A.1). Full-pool 7B answering/revision and the unchanged 3B draft are separate pipelines. CoRM's released text-only DeBERTa critic uses its native 256-token cap [21].

4.3 Training and Evaluation Protocols

Controls share main pools, models, budgets, partitions, features, training, and metrics except investigated factors. Unchanged checkpoints are reused; affected components retrain. Acceptance parameters use identical development splits and search. Held-out confirmation changes only test questions. No dataset-specific online auxiliary features are used.

Scorers share AdamW/input settings (Appendix A.2); gain models follow the question-grouped ridge protocol. Single-seed model runs use 2094; multi-seed studies retain declared seed sets. Matched-input, reader, budget, and 7B-reference controls retain their tested variables. Input/output caps are 4,072/24 for drafts, 2,048/32 for greedy reading, and 8,192/32 for full-pool 7B; runs use two RTX 4090 GPUs.

Evaluation and uncertainty. Normalized complete-answer EM and token F1 maximize over gold aliases; failures stay in denominators. Macro weights datasets equally; differences are percentage points. Paired question-bootstrap CIs use 10,000 draws, seed 2092 (Appendix B.1). Training variation uses sample SD; ensemble disagreement uses population SD. Quality uses full sets; timing uses 250 fixed questions (50/dataset), three rounds after five warm-ups. Requests span available pool to answer, including IPC, excluding retrieval, model loading, warm-ups, and writing. Repeats add no quality samples.

Gain and decision metrics. For held-out main-test questions Q_d^+ with nonempty proposals, reader F1 is normalized to $[0, 1]$, and the corresponding gains lie in $[-1, 1]$. We compute gain-prediction MAE as follows:

$$\mathrm{MAE} = \frac{1}{5}\sum_{d=1}^{5}\frac{1}{|Q_d^+|}\sum_{q\in Q_d^+}\frac{\sum_{S'\in\mathcal{P}(q)}|\mu(S')-\Delta(S')|}{|\mathcal{P}(q)|}. \tag{15}$$

Rejected proposals count. All gain-formulation controls share initial pairs, proposals, and reader outputs. Acceptance/harm counts pool all 4,500 questions in one run; harm strictly decreases reader F1 from the initial pair.

5 Experimental Results

5.1 System Performance and Held-out Confirmation

Table 1 compares nine systems and reports paired intervals for DCR-RAG's gains over each baseline. DCR-RAG reaches 40.28 EM/54.46 F1: +2.82 over SetR [1.91, 3.74], and +2.93 over REARANK [2.04, 3.82]. It leads F1 on four datasets; ERank leads on NQ. Its largest SetR gains are on 2Wiki (6.44) and MuSiQue (3.96). The 8.30-point lead over GainRAG-controlled is a complete-system contrast.

Table 2: Held-out confirmation using main trained checkpoints: EM/F1 (%) on 1,000 disjoint questions each from HotpotQA and 2Wiki. Macro uses these two datasets only.

Route	HotpotQA	2Wiki	Macro
DCR-RAG	65.70/76.10	33.00/50.24	49.35/63.17
SetR-2 (controlled)	65.50/75.76	28.70/45.76	47.10/60.76
Main scorer, independent Top-2	64.20/74.82	28.90/45.10	46.55/59.96
BGE + same draft	64.30/74.90	25.80/41.96	45.05/58.43

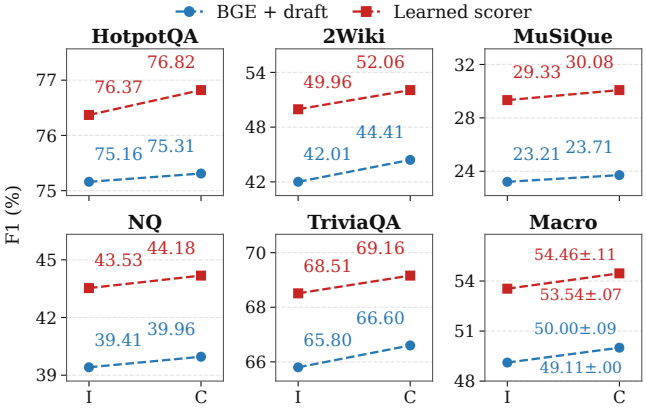


Figure 2: Final-checkpoint scorer × selection. I/C: independent/conditional. Dataset F1 uses seed 2094; macro labels are three-seed means ± sample SD.

Table 2 reuses main checkpoints on disjoint HotpotQA/2Wiki questions. DCR-RAG exceeds SetR by 2.41 macro F1 [1.36, 3.47], mainly from 2Wiki. Full and independent Top-2 routes share the main scorer; their 3.21-point gap concerns this two-dataset held-out population. Appendix B.1 gives further confirmation statistics and CoRM transfer results.

5.2 Scorer Learning and Teacher Supervision

Figure 2 fixes questions, pools, drafts, reader, pair predictor, and budget; each checkpoint serves both selection routes. Scorer learning adds 4.43 F1 [3.51, 5.35] under independent selection; the complete selector adds 0.92 [0.39, 1.45] with the final scorer fixed. Interaction is 0.03 [−0.48, 0.54]. Shared drafts distinguish scorer learning from draft access.

With pool drafts fixed, hybrid supervision adds 2.94/1.77 full-system macro F1 over BGE-only/deletion-only labels (Table 3a). No-backfill lowers independent/full F1 by 0.04, testing context repacking. The separate eight-passage, 7B-reference teacher control shows no clear advantage for richer heads (Appendix B.2).

5.3 What Information Does the Draft Add?

Retrained no-draft controls give +1.89 [1.22, 2.57] macro F1 (54.46 versus 52.57), positive on all five datasets (Table 3b). Pool drafts add 1.62 over question-only drafts (panel a). These test draft access and pool conditioning, respectively.

The length-matched draft/expansion control retrains both arms with identical Necessity labels, initialization, optimization, reader, and questions; only auxiliary content differs (Appendix B.2). This separate fixed-supervision protocol tests draft content, rather than the full Hybrid teacher configuration. Drafts score 50.95 versus 49.39: +1.56 [0.98, 2.14], on 4,332 retained questions, positive per dataset (Table 4).

Table 3: Input and teacher ablations (F1, %): (a) macro results; (b) retrained no-draft control by dataset. ΔF_1: full minus no draft. Point estimates; bold: best macro in (a).

(a) Draft source and teacher: macro F1

Configuration	Independent	Full
Pool draft + hybrid teacher	**53.54**	**54.46**
Question-only draft + hybrid teacher	51.96	52.84
Pool draft + BGE-only teacher	50.73	51.52
Pool draft + deletion-only teacher	51.82	52.69
Pool draft + hybrid, no backfill	53.50	54.42

(b) Final-configuration draft access

Dataset	No draft	Full draft	ΔF_1
HotpotQA	74.63	76.82	+2.19
2Wiki	49.44	52.06	+2.62
MuSiQue	28.64	30.08	+1.44
NQ	43.02	44.18	+1.16
TriviaQA	67.12	69.16	+2.04
Macro	52.57	54.46	+1.89

Table 4: Exactly length-matched draft/expansion control: retrained arms and retained questions. Per-dataset and macro 95% confidence intervals use seed-averaged paired question bootstrap.

Dataset	n	Draft	Query expansion	ΔF_1	95% CI
HotpotQA	968	75.34	72.73	+2.61	[1.24, 3.98]
2Wiki	971	47.18	45.30	+1.88	[0.48, 3.29]
MuSiQue	473	24.45	22.93	+1.52	[0.08, 2.96]
NQ	958	41.47	40.41	+1.06	[0.06, 2.08]
TriviaQA	962	66.32	65.58	+0.74	[0.05, 1.44]
Macro	–	50.95	49.39	+1.56	[0.98, 2.14]

Table 5: Fixed-scorer policies: macro F1 (%) and Direct Gain (54.46) minus policy, with paired 95% CIs. Common inputs, budgets, and development tuning.

Policy	F1	Direct Gain minus policy
Independent Top-2	53.54	+0.92 [0.39, 1.45]
MMR-style	53.63	+0.83 [0.21, 1.45]
Coverage	53.75	+0.71 [0.10, 1.32]
Combined	53.87	+0.59 [0.02, 1.20]

5.4 Replacement Decisions and Reliability

Unless otherwise specified, DCR-RAG uses the original proposal-only predictor, termed Direct Gain here.

Fixed-scorer selection baselines. With drafts, trained scorer, pools, reader, Top-2 budget, questions, and development tuning shared, Direct Gain exceeds MMR/coverage/combined by 0.83/0.71/0.59 F1 points; all paired 95% intervals exclude zero (Table 5). Heuristics use no reader-gain labels (Appendix B.3).

Table 6: Gain formulations: MAE over all valid proposals on normalized F1 gains, main-test macro F1 (%), and Direct Gain minus alternative with paired 95% CIs. Initial pairs, proposals, and reader outputs are shared.

Configuration	MAE	F1	ΔF_1	95% CI
Direct Gain	**0.164**	**54.46**	—	—
Baseline-aware (three blocks)	0.174	54.20	+0.26	[0.02, 0.50]
Baseline-aware (two blocks)	0.177	54.10	+0.36	[0.04, 0.68]
Difference-feature gain	0.180	54.02	+0.44	[0.05, 0.83]
Shared absolute-F1 difference	0.188	53.78	+0.68	[0.15, 1.21]

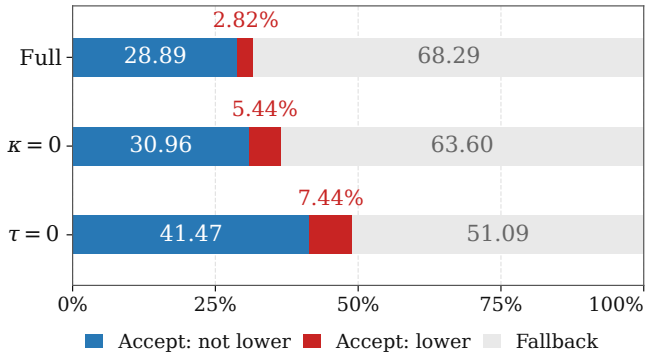


Figure 3: Acceptance outcomes on all 4,500 questions (share denominator). Nondecreasing accepts include ties.

Gain formulation and initial-pair dependence. Table 6 compares four ways to incorporate initial-pair information, keeping initial pairs, proposals, and reader outputs fixed.

Direct Gain has lower MAE and higher F1; all paired F1 intervals exclude zero. Removing the dependent difference block does not improve F1. At 31.67–31.82% acceptance, three alternatives have 10.91–13.06% harm/accepted, versus 8.90% at 31.71% for Direct Gain (Table 10b).

Frozen gain models, preprocessing, and acceptance retain 0.25–0.43 F1 under changed initial policies, with positive CIs and MAE 0.178–0.190 (Table 10a). Transfer depends on initial pairs and proposals; lower MAE can accompany higher harm/accepted across initial-F1 strata (Table 10c).

Replacing the ensemble decision rule with Single Ridge yields 0.181 MAE/54.08 F1, 0.38 below the full rule; zero-gain MAE is 0.205. Appendix B.3 gives control protocols and within-policy gain intervals.

Harm strictly decreases F1 from the initial pair. Full-rule acceptance is 1,427/4,500 (31.71%): 127 harmful (8.90% of accepts; 2.82% of all). For $\kappa = 0/\tau = 0$, acceptance/harm-all is 36.40/5.44% (245 harmful) and 48.91/7.44% (Figure 3); full-rule gains are 0.28 F1 [0.04, 0.52]/0.54 [0.22, 0.86].

Disagreement-aware acceptance. Mean-only shares the ensemble means, sets $\kappa = 0$, and matches acceptance using development-selected thresholds frozen for testing. Full/mean-only accepts are 1,427/1,431 (31.71/31.80%), including 488/444 beneficial and 127/210 harmful decisions. Harm/accepted falls from 14.68% to 8.90%; macro F1 rises from 53.83 to 54.46: +0.63 [0.04, 1.22], or 0.63 ± 0.02 across three seeds (sample SD). Full/mean-only gains over independent Top-2 are 0.92/0.29; their 0.63 contrast tests disagreement-aware decisions. Acceptance sweeps fix each rule's κ, tune dataset-specific thresholds on development data, and freeze them for testing (Table 11). The full rule has higher F1 and lower harm/accepted at the reported targets; disagreement remains uncalibrated.

Table 7: Reader transfer: three-seed macro F1 means ± sample SD, draft-minus-no-draft effects, and paired 95% intervals. Each arm reuses its selected IDs across readers.

Reader	No draft	Draft conditioned	ΔF_1	95% CI
Qwen2.5-1.5B	43.93 ± 0.08	45.35 ± 0.10	+1.42 ± 0.06	[0.61, 2.23]
Qwen2.5-3B	46.89 ± 0.07	48.60 ± 0.09	+1.71 ± 0.05	[0.91, 2.51]
Qwen2.5-7B	52.57 ± 0.07	54.46 ± 0.11	+1.89 ± 0.05	[1.22, 2.57]
Qwen2.5-14B	51.13 ± 0.09	53.14 ± 0.08	+2.01 ± 0.06	[1.14, 2.88]
Llama-3.1-8B	49.05 ± 0.08	50.87 ± 0.10	+1.82 ± 0.05	[1.01, 2.63]
Mistral-7B-v0.3	44.24 ± 0.11	45.60 ± 0.10	+1.36 ± 0.07	[0.46, 2.25]
InternLM3-8B	47.78 ± 0.09	49.62 ± 0.08	+1.84 ± 0.06	[0.96, 2.71]

6 Robustness and Budget Trade-offs

6.1 Passage Use and Draft Robustness

Support recall averages per-question fractions of annotated supports; complete coverage requires all supports. DCR-RAG raises HotpotQA/2Wiki recall from 90.90/81.20% to 94.30/86.55% and complete coverage from 82.30/62.60% to 89.20/73.30%. MuSiQue recall rises from 45.38 to 48.25% (+2.87, [1.33, 4.35]) and complete coverage from 2.60 to 3.00%; reader-visible values are 48.15/3.00% at $k = 2$. On four-hop MuSiQue, recall rises at position four and falls at position one; complete coverage is unchanged.

Equal-length support/nonsupport interventions fix question, other passage, and decoding. On 162 jointly eligible MuSiQue questions, support/nonsupport replacement changes F1 by −15.21/−1.33 points; the paired difference is −13.88 [−17.12, −10.64], supporting reader dependence on selected passages. Natural correct/wrong drafts have 4.12/6.76% support exclusion from the initial pair. Matched entity/relation edits fix other draft text, pools, scorer, and reader; on 826/748 questions, strict-EM failures rise 6.54 [4.11, 8.95]/8.82 [5.99, 11.65] points. These edits test error propagation; natural strata are observational (Appendix C).

6.2 Transfer and Training Stability

LODO retains the target-trained scorer; gain models, preprocessing, and acceptance are fitted without target-domain gain-training or development data and frozen for evaluation. It reaches 54.18 macro F1: +0.64 [0.28, 1.00] over independent selection and −0.28 [−0.52, −0.04] versus target fitting, with gains on all targets (Appendix C).

Across seven readers, draft conditioning adds 1.36–2.01 F1 (mean 1.72 [1.08, 2.35], joint question–seed; Table 7), reusing each arm/seed's selected IDs and withholding drafts from readers. Scorer/predictor retraining at 2094–2096 yields gains of 4.43 ± 0.07 for scorer learning under independent selection and 0.92 ± 0.04 for selection with the final scorer (sample SD). Joint question–seed CIs are [3.05, 5.78]/[0.28, 1.56]; Table 13 gives seed-wise scores.

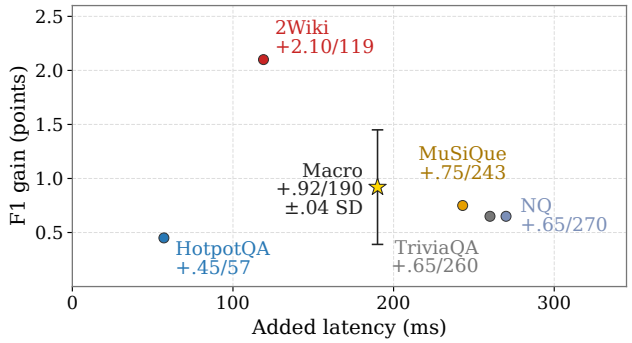


Figure 4: Fixed-scorer F1 gain versus added complete-request time. Full quality sets; 250 timing questions. Black bar: paired 95% CI; SD: three training seeds.

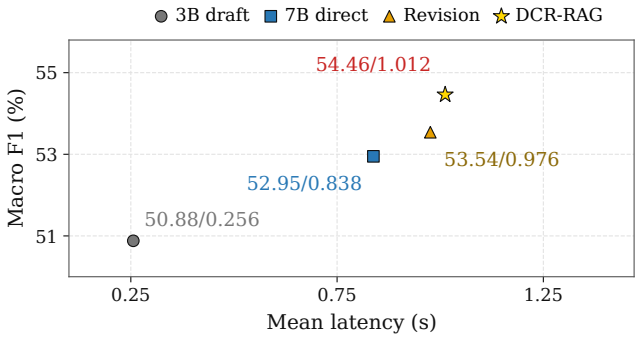


Figure 5: Full-pool pipelines: macro F1 / mean request seconds. Quality: 4,500 questions; timing: 250. Full-pool 7B uses 8,192 input/32 output tokens; DCR-RAG reads Top-2. Point estimates.

6.3 Quality, Cost, and Passage Budgets

With the scorer fixed, the complete selector adds 0.190 request seconds [0.176, 0.204] (Figure 4).

DCR-RAG gains 1.51 F1 points over direct full-pool answering for 0.174 seconds (0.838 to 1.012, approximately 21%), and 0.92 points over revision for 0.036 seconds [0.018, 0.054] (Figure 5).

Search scope. Eight slots, sixteen slots, and exhaustive prediction evaluate 7.4/14.1/51.3 deduplicated pairs on average (Figure 6). With models and reader fixed, sixteen slots/exhaustive search add 0.15/0.28 F1 for 0.055/0.170 seconds. Eight slots contain an exhaustive risk-score maximizer on 92.40% of questions, counting ties. The 61.85 observed-reader-F1 oracle uses gold answers offline, distinct from maximizing predicted risk-adjusted gain.

Passage budget. On all 500 MuSiQue questions, $k = 1$ uses the top scorer, $k = 2$ the selected pair, and $k = 3, 4$ append top remaining scores without retraining or retuning. BGE order and the 2,048-token reader cap stay fixed. Two to four passages adds 3.13 F1 for 0.396 seconds; truncation limits visible support (Figure 7). With two passages, DCR-RAG exceeds SetR/BGE by 3.96/10.21 F1; with four, by 3.86/7.81 (Table 8). The advantage persists at four passages, although the fourth adds a smaller gain.

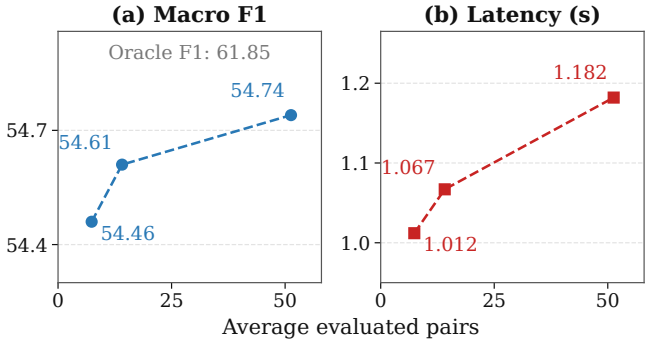


Figure 6: Pair-search scope: (a) main-test macro F1, (b) timing-subset mean seconds. The gold observed-F1 oracle (61.85) is offline and differs from the learned risk-score maximum.

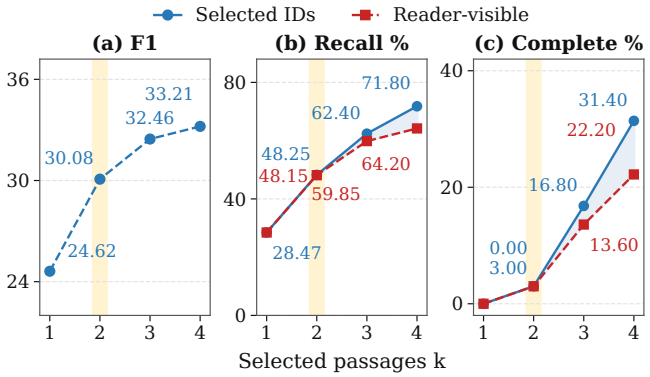


Figure 7: MuSiQue passage budget: (a) F1, (b) support recall, (c) complete coverage. Solid: selected IDs; dashed: reader-visible support. Reader cap: 2,048 tokens; highlighted $k = 2$ is standard.

Table 8: MuSiQue F1 (%), $n = 500$, common reader and matched budgets. Point estimates.

Method	$k = 2$	$k = 4$
BGE Top-k	19.87	25.40
SetR-k	26.12	29.35
DCR-RAG	**30.08**	**33.21**

These results favor compact passage sets under constrained budgets, without a general speed advantage over full-context generation. Appendix C gives request distributions and coverage-counting details.

7 Conclusion

DCR-RAG conditions passage scoring on candidate-pool drafts and accepts local replacements through predicted reader gain, disagreement, and a threshold. With shared pools, reader, and Top-2 passages, it reaches 54.46 macro F1, 2.82 points above SetR. Matched controls distinguish draft and scorer effects; near-matched acceptance supports the decision rule. Passage-use and cost analyses bound its use (Appendix D).

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A Implementation and Reproducibility

A.1 Models, Inputs, and Feature Processing

Execution settings. The draft model uses vLLM context 8,192 and memory allocation 0.45; input/output caps appear in Section 4.3. Full-pool revision also receives the draft. GainRAG-controlled fine-tunes BGE-reranker-base on 100 questions/dataset (Train100, seed 2094), without pseudo-passages; the common Qwen2.5-7B-Instruct reads Top-2 from frozen BGE pools. The anonymous repository identifies the SetR variant and trained weights, GainRAG checkpoints, prompts, data IDs, seeds, and execution settings.

Controlled SetR configuration. SetR fills its 10,240-token Llama-3.1-8B-Instruct selector context in BGE order and outputs Top-2 in that order, using the common pools and reader. The repository records decoding, validation, and deterministic handling of non-conforming responses.

Feature processing. Student and BGE scores are min–max normalized per pool; constant pools return zero. Pairs use candidate indices fixed before student reranking. Lexical features use Unicode NFD, case folding, and Unicode `\w+` tokenization, retaining articles/stopwords. Draft match requires a contiguous occurrence of the complete normalized draft token sequence in passage title and text; empty drafts never match. Passage overlap is $|T_u \cap T_v|/\max(1, |T_u \cup T_v|)$.

Draft-match nonzero rates over all evaluated candidate passages are 0.90, 0.65, 0.25, 0.55, and 1.10% for HotpotQA, 2Wiki, MuSiQue, NQ, and TriviaQA. Joint draft-conditioned scoring requires no exact match.

A.2 Teacher Construction and Training

Main teacher. For Eq. 3, $\ell(\hat{y} \mid q, S)$ is Qwen2.5-3B-Instruct's mean log probability over the fixed draft's answer tokens, excluding prompt, padding, and an additionally appended EOS. The deletion response is

$$N_i = \max\{0, \ell(\hat{y} \mid q, D) - \ell(\hat{y} \mid q, D \setminus \{d_i\})\}. \tag{16}$$

Deletion preserves passage order and IDs and rebuilds the prompt under the 4,072-token cap, exposing previously truncated text; the draft is not regenerated. Frozen BGE scores the question against passage title and text; constant pools normalize to zero. The quantile c is floored at 10^{-8}, or set to one without positive responses. Empty drafts set $N_i = 0$. Ties follow BGE order then ID; nonfinite utilities are excluded, with BGE Top-2 fallback if fewer than two remain.

Scorer and gain training. Scorers are trained separately on 100 questions per dataset, drawn from HotpotQA distractor validation, 2Wiki train, MuSiQue-Ans train, and the E5 retrieval files `train_1w.json` for NQ/TriviaQA. Scorer and gain-training questions are disjoint; ID/normalized-question checks exclude evaluation questions.

AdamW updates all scorer parameters for three epochs: learning rate 10^{-6}, linear schedule, warm-up 0.1, weight decay 0.01, one question group per device, and accumulation four. Training uses FP16 and gradient checkpointing; the final epoch supplies the checkpoint. The question–draft query and passage are pretruncated to 200/464 tokens, then jointly encoded with a 512-token cap and `only_second` truncation. No separate draft cap is imposed within the query. Groups contain at least three candidates; preferences are averaged within group.

Gain-training folds are

$$f(q) = \text{int}[\text{SHA256}(s \| \text{dataset}(q) \| \text{id}(q))] \bmod 5, \tag{17}$$

with frozen s. Each ridge member fits its own standardizer and weights on four folds, $\alpha = 10$, with an intercept and constant-feature scale one. Main seed is 2094; stability seeds 2094–2096 retrain scorer and predictors with generators, reader, pools, splits, prompts, and scoring fixed.

B Controlled Evaluation and Diagnostics

B.1 Evaluation and System Comparisons

Evaluation and uncertainty. Complete answers maximize over gold aliases without post-hoc extraction; HotpotQA uses official yes/no/noanswer mismatch rules. The empty-normalized NQ alias and generation failures remain in denominators. Pointwise bootstrap intervals resample questions within dataset with route pairing, then macro-average. Seed-averaged intervals first average same-question effects; joint intervals resample questions and seeds with pairing.

System comparisons. CoRM transfer uses the same 2,500 HotpotQA/2Wiki/MuSiQue questions and BGE display order. CoRM/DCR-RAG F1 is 66.45/76.82, 31.67/52.06, and 25.46/30.08; paired gain intervals are [8.05, 12.70], [17.95, 22.81], and [1.30, 7.95]. Its three-dataset macro gain is 11.79, approximately [10.22, 13.36].

Confirmation reuses main checkpoints and frozen thresholds (Table 2). DCR-RAG minus SetR F1 is +0.34 [−1.11, 1.79]/+4.48 [2.95, 6.01] on HotpotQA/2Wiki. The main-scorer independent Top-2 gap is 3.21 [2.15, 4.28]; BGE+same-draft gap is 4.74 [3.69, 5.80].

B.2 Scorer Supervision and Matched Inputs

Truncation backfill. No-backfill deletes only from the originally truncated context, admitting no new text. We recompute teacher preferences, retrain scorer and gain predictors on their training splits, and select acceptance parameters on development data, with other settings fixed (Table 3a). It tests context repacking, not incorrect-draft bias.

7B-reference teacher control. On each of HotpotQA/2Wiki, the independent 7B reference/draft control matches 100 training questions, eight-passage groups, initialization, preference loss, candidate order, Top-2 output, and optimization/truncation (Appendix A.2) across teachers; models are frozen before testing.

With mean reference log likelihood ℓ over nonpadding answer tokens, Scalar uses $U_i^{\text{scalar}} = \ell(a_q^* \mid q, \{d_i\}) - \ell(a_q^* \mid q, \varnothing)$ for a fixed gold alias; Necessity uses Eq. 16 with the non-oracle draft and 7B reference. The null term is constant within question; deletion can expose text after truncation. Structured anchors normalized consensus from BGE Top-3/Top-2 and MMR Top-2. Conflict-minus-anchor margin $m(S)$ gives $H_i = \max\{0, m(S \cup \{d_i\}) - m(S)\}$, with $d_i \notin S$. A type-preserving minimal edit $s \to t$ defines $m_{t,s}(d) = \ell(t \mid q, d) - \ell(s \mid q, d)$ and $C_i = \max\{0, m_{t,s}(d_i^{\text{cf}}) - m_{t,s}(d_i)\}$. Consensus, source uniqueness, insertion, edit validity, and direction set masks

Table 9: Independent 7B-reference teacher control with eight-passage groups: F1 (%), means ± sample SD over seeds 2094–2098; 1,000 held-out questions per dataset.

Teacher / anchor	HotpotQA	2Wiki	Macro
BGE anchor	71.82	56.74	64.28
No-draft student	72.91	58.06	65.48
Scalar	74.09 ± 0.06	62.24 ± 0.04	68.17 ± 0.04
Necessity	74.13 ± 0.07	62.25 ± 0.06	68.19 ± 0.05
Structured	74.15 ± 0.05	62.20 ± 0.06	68.18 ± 0.05

m_i^H, m_i^C. Invalid/unavailable signals and masks are zero; valid zeros retain masks one.

The semantic prior U_i^{sem} is frozen BGE's raw question/title-plus-text score. With pool min–max z_D (constant pool zero) and training-standardization z_{tr} (constant-feature scale one),

$$z_i = [z_D(U_i^{\text{sem}}), z_{\text{tr}}(N_i), -m_i^H z_{\text{tr}}(H_i), \\ m_i^C z_{\text{tr}}(C_i), m_i^H, m_i^C]. \tag{18}$$

A training-only linear head supplies Top-2 selected–unselected labels; likelihoods are offline. Ties use BGE order/ID, with BGE Top-2 fallback if fewer than two finite scores remain. Necessity–Scalar, Structured–Scalar, and Structured–Necessity give paired differences +0.02 [−0.07, 0.11], +0.01 [−0.08, 0.10], and −0.01 [−0.10, 0.08] (Table 9); respective Top-2 agreement/preference conflict is 93.1/5.8%, 92.7/6.2%, and 94.4/4.9%.

Matched drafts and query expansion. Both arms greedily generate Qwen2.5-3B text from common candidate context (128-token cap). Whole-word prefixes match auxiliary/complete-query counts with caps 64/200, without filler. Matching retains 4,332 test questions and 95/97 HotpotQA/2Wiki training questions (mean auxiliary lengths 13.20/11.58). Matched arms retrain at seeds 2094–2096. All retained inputs pass prefix, passage-ID, and nonpadding-length checks. Intervals average same-question outcomes across seeds before paired question resampling. The macro 95% confidence interval is computed by equally averaging the five dataset-level gains within each bootstrap replicate (Table 4). Draft/expansion support recall is 92.23/90.50% on HotpotQA and 88.79/87.01% on 2Wiki.

Matched-input confirmation. Table 2 reuses main checkpoints (2Wiki F1 50.24); separate input confirmation retrains matched arms. Three-seed no-draft/full-draft F1 is 71.84/74.52 on HotpotQA and 38.71/42.02 on 2Wiki; gains 2.68 [1.43, 3.91]/3.31 [1.86, 4.74]. Macro is 55.28/58.27, +2.99 [2.07, 3.91], joint CI [1.84, 4.12].

B.3 Selection and Decision Controls

Fixed-scorer heuristics. Let $p_i = z_D(s_i)$, $A = (p_u + p_v)/2$, and J be passage Jaccard overlap. Using the coverage and residual features, set $C = \text{cover}(u, v)$ and $R = [\text{residual}(v \mid u) + \text{residual}(u \mid v)]/2$. Pair scores are

$$H_{\text{MMR}} = A - \lambda J, \qquad H_{\text{Cov}} = A + \beta C, \tag{19}$$

$$H_{\text{Comb}} = A + \beta C + \gamma R - \delta J. \tag{20}$$

Maximize H over $\{S_0\} \cup \mathcal{P}(q)$: MMR-style compares pairs jointly rather than fixing the first passage. Ties keep S_0, then BGE order/ID.

Table 10: Selection diagnostics on 4,500 questions: (a) frozen-model transfer across initial policies; (b) formulations at comparable acceptance; (c) offline initial-F1 strata. CI: (a) after-minus-initial F1; excess: (b) harm/accepted minus Direct Gain. F1 in a and all rates (%); initial-F1 bins use $[0, 1]$.

(a) Initial-policy transfer: frozen gain-trained models

Policy	Initial F1	After F1	Gain	MAE	95% CI
Top-2	53.54	54.46	+0.92	0.164	[0.39, 1.45]
MMR-style	53.63	54.06	+0.43	0.178	[0.12, 0.74]
Coverage	53.75	54.10	+0.35	0.184	[0.07, 0.63]
Combined	53.87	54.12	+0.25	0.190	[0.02, 0.48]

(b) Formulations at comparable acceptance rates

Method	Accepted (%)	Harm/acc. (%)	Excess (points)
Direct Gain (full selector)	31.71	**8.90**	—
Baseline-aware concatenation	31.78	10.91	+2.01
Difference-feature gain	31.67	12.00	+3.10
Shared absolute-F1 difference	31.82	13.06	+4.16

(c) Direct Gain by offline initial-F1 stratum

Initial F1	n	MAE	Accept n	Accept (%)	Harm n	Harm/acc. (%)
[0, 0.25)	1,000	0.203	310	31.00	8	2.58
[0.25, 0.50)	900	0.179	280	31.11	16	5.71
[0.50, 0.75)	800	0.157	255	31.88	27	10.59
[0.75, 1.00]	1,800	0.137	582	32.33	76	13.06
Total	4,500	—	1,427	31.71	127	8.90

Note: MAE uses valid proposals within each initial-F1 bin; formal dataset-macro MAE follows Eq. 15 (Table 6).

Development macro F1 selects from $\{0, 0.25, 0.5, 0.75, 1\}$: five settings each for MMR and coverage, 125 for combined, frozen for testing. No reader-gain labels are used.

Reference-pair formulations. All gain-formulation controls share training questions, proposals, reader outputs, and development threshold selection. Difference-feature gain regresses $\Delta(S')$ on $x_{S'} - x_{S_0}$ with the same five-fold ridge protocol. Shared absolute-F1 difference trains a shared ridge predictor h on absolute reader F1 and predicts $h(x_{S'}) - h(x_{S_0})$. Baseline-aware concatenates:

$$x_{S'}^{\text{BA}} = [x_{S'}; x_{S_0}; x_{S'} - x_{S_0}]. \tag{21}$$

The last block is linearly dependent on the first two, changing ridge regularization as well as representation. The two-block control uses $x_{S'}^{\text{2B}} = [x_{S'}; x_{S_0}]$, omitting that difference block (Table 6).

Single Ridge uses the same gain-training data, features, and regularization, with one predictor and no disagreement penalty; other components remain fixed.

Table 10a separates changed-initial-policy transfer from the fixed-initial-pair formulations in panel b; panel c conditions Direct Gain outcomes on initial F1.

Acceptance outcomes. Full/mean-only unchanged counts are 812/777; mean positive/negative F1 changes are +0.115/−0.109 and +0.093/−0.135.

For full/$\kappa = 0/\tau = 0$, harm/accepted counts are 127/1,427, 245/1,638, and 335/2,201 (8.90/14.96/15.22%). Full at 50% retains

Table 11: Measured test acceptance at development targets, using main κ (full) or zero (mean-only) and development-selected, frozen thresholds per dataset. H/A: harm/accepted (%, rounded); F1: macro (%).

	Full rule			Mean-only		
	Accepted			Accepted		
Target	n (%)	H/A	F1	n (%)	H/A	F1
20%	906 (20.13)	6.7	54.24	898 (19.96)	9.8	53.80
25%	1,132 (25.16)	7.8	54.36	1,119 (24.87)	12.5	53.85
31.7%	1,427 (31.71)	8.9	54.46	1,431 (31.80)	14.7	53.83
40%	1,816 (40.36)	11.3	54.35	1,805 (40.11)	18.0	53.77
50%	2,241 (49.80)	15.4	54.18	2,262 (50.27)	22.0	53.64

Table 12: MuSiQue subset support recall (%) by decomposition position with Top-2 passages. Tuples follow hop order.

Type	n	BGE Top-2	DCR-RAG
2-hop	24	(87.50, 54.17)	(87.50, 62.50)
3-hop	324	(76.23, 29.63, 41.98)	(74.69, 30.25, 50.62)
4-hop	152	(71.71, 13.82, 23.03, 23.68)	(63.82, 17.76, 25.00, 38.82)

main κ, as for $\tau = 0$: 345/2,241 harm/accepted (15.4%; 7.67% harm/all). Mean-only harm/all is 4.67%.

C Robustness, Transfer, and Cost Details

Support coverage and reader edits. Complete-support counts (BGE, BGE+draft I/C, final I/C): 823/850/861/874/892 (HotpotQA), 626/669/690/708/733 (2Wiki), each of 1,000. MuSiQue BGE Top-2/DCR-RAG cover 13/15 of 500 (2.60/3.00%), all two-hop. Truncation retains all 15 for DCR-RAG; recall falls from 48.25 to 48.15%. Independent recall/complete coverage: 93.35/87.40% (HotpotQA), 85.10/70.80% (2Wiki), below full DCR-RAG.

Reader edits fix the question, other passage, and greedy decoding; natural nonsupport donors match the full chat-prompt token count. MuSiQue support/nonsupport replacement ($n = 448/187$) changes F1 by -15.53 $[-18.44, -12.67]/-1.70$ $[-4.01, 0.50]$; deletion ($n = 477/216$) gives -13.34 $[-16.01, -10.65]/-2.35$ $[-5.07, 0.49]$. Paired replacement uses 162 jointly eligible questions.

Support/nonsupport replacement changes F1 by $-14.12/-1.96$ on HotpotQA and $-16.56/-1.56$ on 2Wiki.

Table 12 gives MuSiQue's decomposition-position recall; saved HotpotQA/2Wiki inputs lack corresponding hop-position definitions. On four-hop questions, fourth-position recall improves 15.13 points $[9.21, 21.05]$, while first-position recall declines 7.89 $[2.63, 13.82]$; complete-coverage change is 0.00 $[0.00, 0.00]$.

Draft perturbations. Strict draft EM gives correct/wrong counts 653/347 (HotpotQA) and 341/659 (2Wiki); verbose or unlisted aliases may fail. Exclusion removes support present in independent Top-2. Natural-correct/wrong exclusion is 41/994 and 68/1,006 (CIs $[3.05, 5.55]/[5.37, 8.48]$%). Correct-draft/wrong-final cases are 49/994 (4.93%, $[3.75, 6.46]$). Entity/relation edits replay natural drafts: exclusion is 81/826 (9.81%, $[7.96, 12.02]$)/88/748 (11.76%, $[9.65, 14.27]$), with 97/826 and 104/748 final failures. Strict-EM failure rises from 5.21/5.08% to 11.74/13.90%.

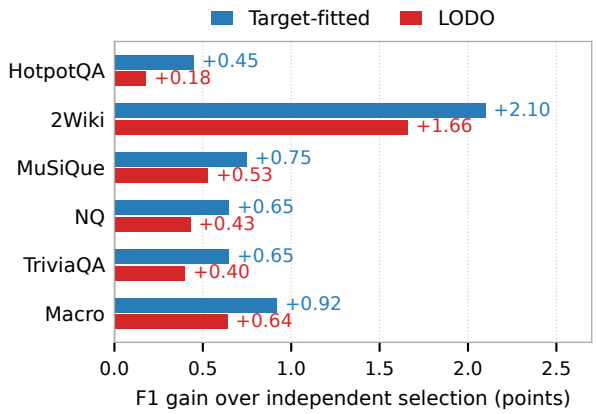


Figure 8: Gain/acceptance transfer retaining target-trained scorers. Paired LODO-gain CI: $[0.28, 1.00]$.

Table 13: Seed-wise macro F1 and effects; last column: three-run mean ± sample SD. BGE independent is fixed.

Route or effect	2094	2095	2096	Mean ± SD
BGE + draft, independent	49.11	49.11	49.11	49.11 ± 0.00
BGE + draft, conditional	50.00	49.91	50.09	50.00 ± 0.09
Final scorer, independent	53.54	53.47	53.61	53.54 ± 0.07
Full DCR-RAG	54.46	54.35	54.57	54.46 ± 0.11
Mean-only (matched)	53.83	53.70	53.96	53.83 ± 0.13
Full minus mean-only	+0.63	+0.65	+0.61	+0.63 ± 0.02
Scorer effect, independent	+4.43	+4.36	+4.50	+4.43 ± 0.07
Scorer effect, conditional	+4.46	+4.44	+4.48	+4.46 ± 0.02
Selector effect, BGE	+0.89	+0.80	+0.98	+0.89 ± 0.09
Selector effect, final scorer	+0.92	+0.88	+0.96	+0.92 ± 0.04
Interaction	+0.03	+0.08	−0.02	+0.03 ± 0.05

Transfer and cost protocols. LODO gains appear in Figure 8. On two RTX 4090 GPUs (2Wiki: GPU1), CUDA-synchronized request timing includes IPC and CPU loading. Independent Top-2/single Ridge/full selector/baseline-aware take 0.822/1.008/1.012/1.016 seconds. Timing-subset DCR-RAG minus BGE is 5.15 F1/0.613 seconds, distinct from main-test quality. Eight slots cover the exhaustive risk maximum on 4,158/4,500 questions (ties included). DCR-RAG requests take 1.012/0.986/1.148 seconds (mean, median, and 95th percentile); full-pool revision takes 0.976/0.949/1.105.

At $k = 1$, MuSiQue hits support on 450/500 questions (22/292/136 at two/three/four hops); mean recall $(22/2 + 292/3 + 136/4)/500 = 28.47\%$ differs from hit rate. Passages 3/4 add 2.38/0.75 F1 for 0.186/0.210 seconds. Complete coverage before truncation is 3.00/31.40% at $k = 2/4$; $k = 4$ selected/visible recall is 71.80/64.20%.

D Limitations

Fixed pools leave live retrieval untested; wrong drafts can misguide selection. Local replacements can miss better pairs; budgets and truncation can omit deeper-hop support. Strong initial pairs leave less headroom; lower gain MAE need not imply safer acceptance.